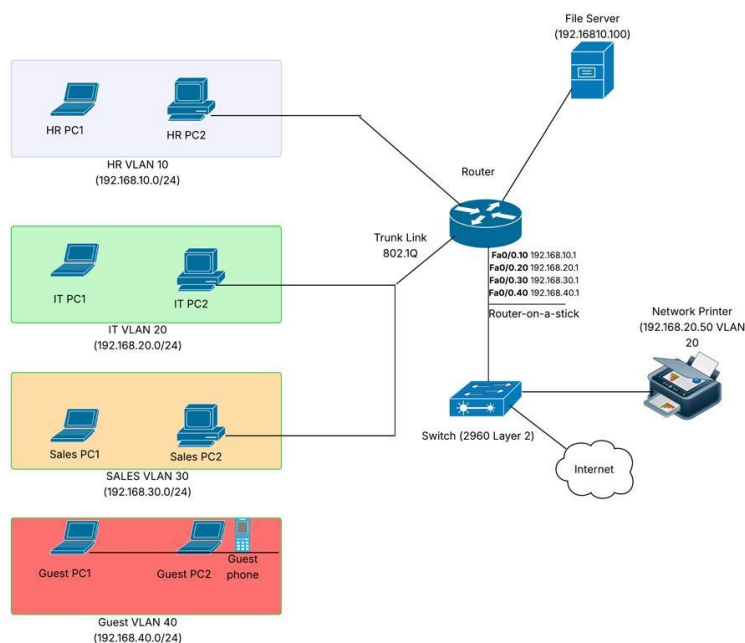


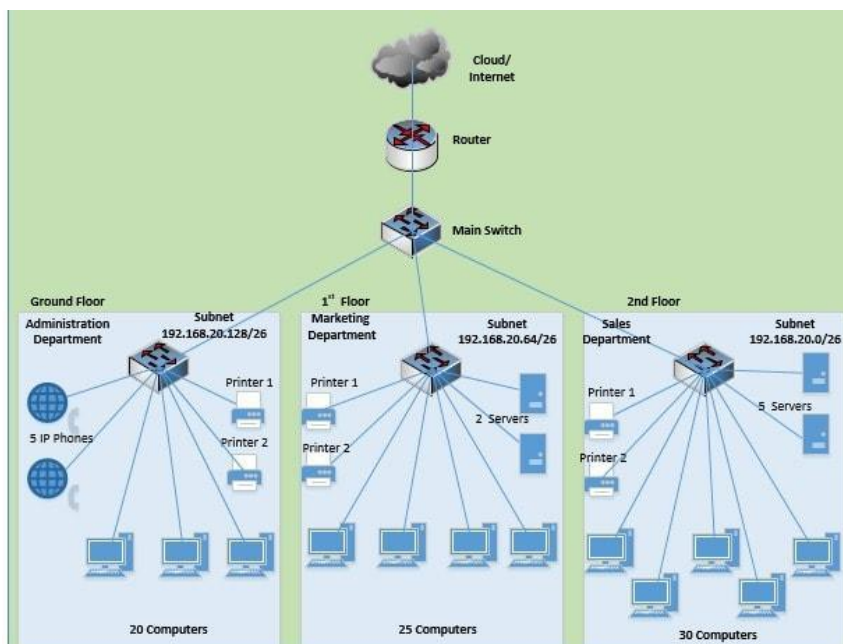
DETAILED DESIGN AND HARDWARE SELECTION

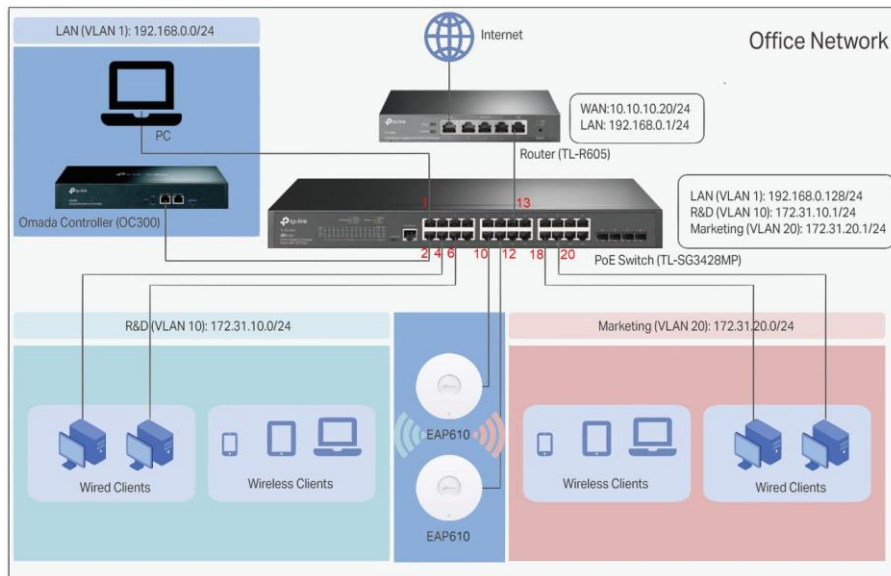
1. Draw a detailed network topology diagram for a small co-working space with 3 departments (e.g., Marketing, Tech, Finance), showing routers, switches, wireless access points, and user devices using any diagram tool like draw.io or Lucidchart.

Small Office Network with VLANs & Inter-VLAN Routing



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2. Create an IP addressing spreadsheet for the above co-working space, assigning unique subnets and VLAN IDs to each department, and listing at least 5 device IPs per department.

 Hint: Use private IP ranges (e.g., 192.168.x.x) and make sure there's no overlap between subnets.

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Co-Working Space IP Addressing Plan

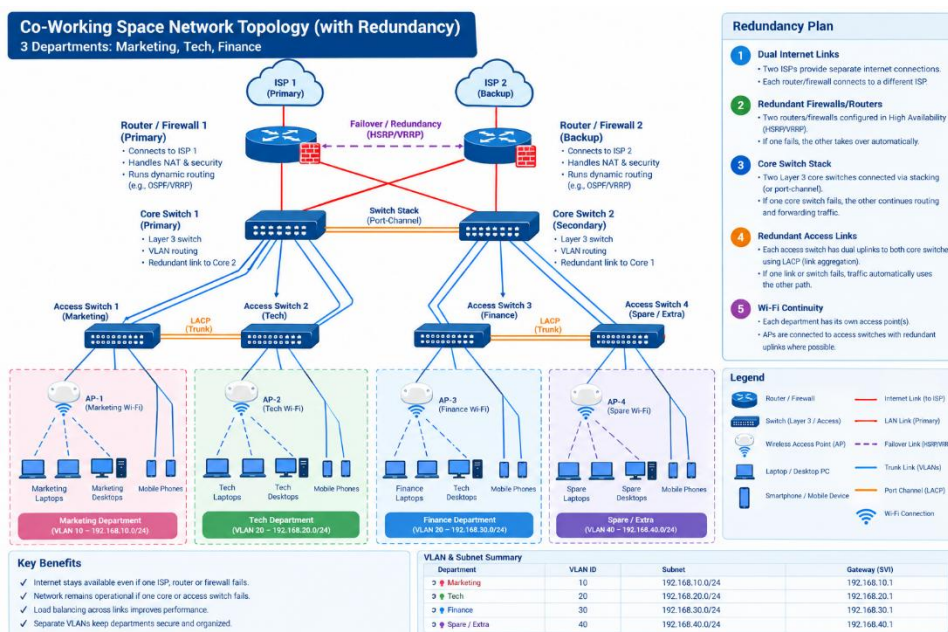
3 Departments: Marketing, Tech, Finance | Private IPv4 Addressing | Non-overlapping /24 subnets

Department	VLAN ID	Subnet	Subnet Mask	Gateway	Device
Marketing	10	192.168.10.0/24	255.255.255.0	192.168.10.1	MKT-PC-01
Marketing	10	192.168.10.0/24	255.255.255.0	192.168.10.1	MKT-PC-02
Marketing	10	192.168.10.0/24	255.255.255.0	192.168.10.1	MKT-LAP-01
Marketing	10	192.168.10.0/24	255.255.255.0	192.168.10.1	MKT-LAP-02
Marketing	10	192.168.10.0/24	255.255.255.0	192.168.10.1	MKT-AP-01
Tech	20	192.168.20.0/24	255.255.255.0	192.168.20.1	TECH-PC-01
Tech	20	192.168.20.0/24	255.255.255.0	192.168.20.1	TECH-PC-02
Tech	20	192.168.20.0/24	255.255.255.0	192.168.20.1	TECH-LAP-01
Tech	20	192.168.20.0/24	255.255.255.0	192.168.20.1	TECH-LAP-02
Tech	20	192.168.20.0/24	255.255.255.0	192.168.20.1	TECH-AP-01
Finance	30	192.168.30.0/24	255.255.255.0	192.168.30.1	FIN-PC-01
Finance	30	192.168.30.0/24	255.255.255.0	192.168.30.1	FIN-PC-02
Finance	30	192.168.30.0/24	255.255.255.0	192.168.30.1	FIN-LAP-01
Finance	30	192.168.30.0/24	255.255.255.0	192.168.30.1	FIN-LAP-02
Finance	30	192.168.30.0/24	255.255.255.0	192.168.30.1	FIN-AP-01

Optional Management Network

Department	VLAN ID	Subnet	Subnet Mask	Gateway	Device
Management	40	192.168.40.0/24	255.255.255.0	192.168.40.1	CORE-SW-01
Management	40	192.168.40.0/24	255.255.255.0	192.168.40.1	AP-MGMT-01
Management	40	192.168.40.0/24	255.255.255.0	192.168.40.1	AP-MGMT-02
Management	40	192.168.40.0/24	255.255.255.0	192.168.40.1	AP-MGMT-03

3. Design a basic redundancy plan for the network that ensures internet connectivity is available even if one main switch fails. Briefly describe your approach and update your topology diagram to include redundancy.



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4. Select suitable network hardware (router, switches, access points) for the co-working space scenario and create a Bill of Materials (BOM) table with device model, quantity, and estimated price (use online prices from Indian retailers like Amazon or Flipkart).

-> For the co-working space, I'd use a **TP-Link Omada-based design** because the components support VLANs, managed switching, PoE for APs, and centralized management. The redundancy plan uses **two core switches**, so if one core switch fails, the access switches can continue through their second uplink.

Bill of Materials (BOM)

#	Hardware	Recommended model	Qty.	Est. unit price	Est. total
1	Router / Multi-WAN Gateway	TP-Link ER605 Omada Gigabit Router	1	₹5,483	₹5,483

#	Hardware	Recommended model	Qty.	Est. unit price	Est. total
2	Core Managed Switch	TP-Link TL-SG3428 JetStream 24-Port Gigabit L2 Managed Switch	2	₹15,200	₹30,400
3	Department PoE Managed Switch	TP-Link TL-SG2210P JetStream 8-Port Gigabit PoE+ Switch	3	₹7,929	₹23,787
4	Wi-Fi 6 Access Point	TP-Link EAP610	3	₹8,013	₹24,039
		Estimated hardware total			₹83,709

Why these devices?

- **ER605:** Provides Gigabit routing, firewall/NAT functions and multi-WAN capability. The current Flipkart listing I found shows approximately **₹5,483**. ([Flipkart](#))
- **TL-SG3428:** A 24-port managed Gigabit switch with VLAN/L2+ capabilities and SFP connectivity. The current Flipkart listing shows about **₹15,200**. Two units provide the core-switch redundancy. ([Flipkart](#))
- **TL-SG2210P:** Has **8 Gigabit PoE+ ports and 2 SFP slots**, making it suitable for each department's access switch and for powering the EAP610 AP. Current Flipkart pricing is around **₹7,929**. ([Flipkart](#))
- **EAP610:** Wi-Fi 6 access point suitable for the Marketing, Tech and Finance areas. The current Flipkart listing shows approximately **₹8,013**. ([Flipkart](#))

TP-Link TL-SG3210 JetStream Managed Switch

₹8,999

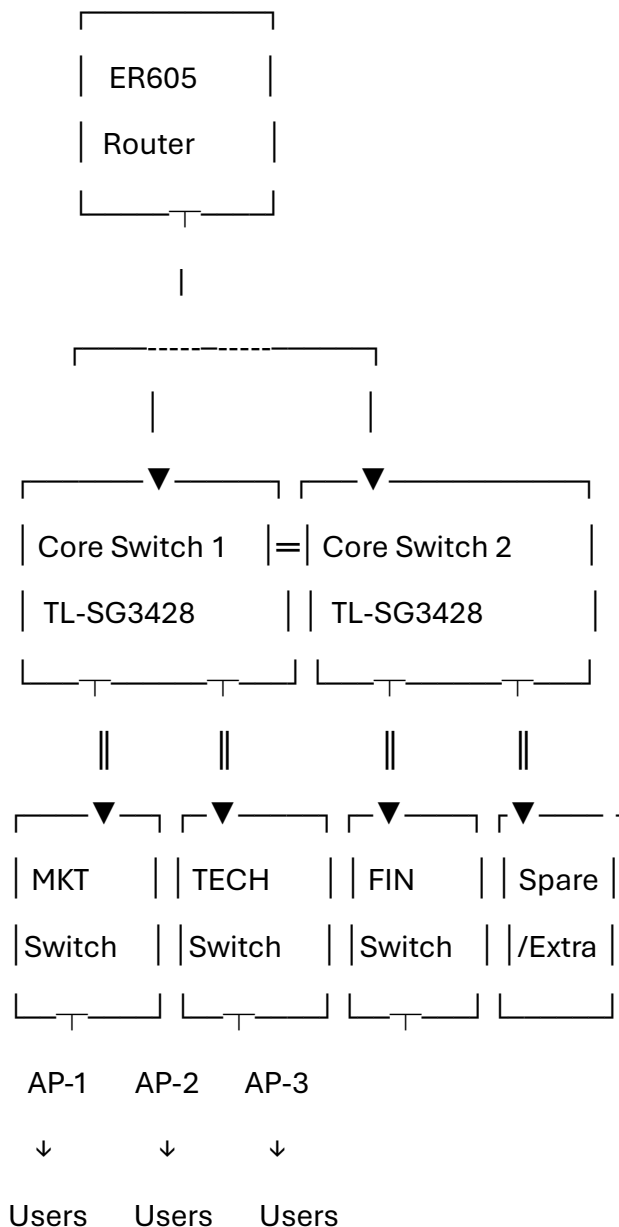
TP-Link T1500G-10PS / TL-SG2210P PoE Switch

₹8,035

Redundancy arrangement

INTERNET

|



Failure scenario: If **Core Switch 1 fails**, the department switches can use their redundant connection to **Core Switch 2**, keeping the Marketing, Tech and Finance networks operational. VLANs 10, 20 and 30 remain separated.

Price note: These are current online Indian retail prices I found during the search, not fixed quotations. Retail prices, seller offers, bank discounts and availability can change. For example, the TL-SG2210P listing showed ₹7,929 on one current Flipkart listing, while another seller/listing showed different pricing. ([Flipkart](#))

Estimated BOM budget: ₹83,709, excluding cabling, rack, UPS, installation and Internet service.

5. Draft a short security policy for the co-working space network covering at least: WiFi password rules, VLAN access restrictions, guest network setup, and

basic firewall guidelines.
Hint: Think about how Zomato or Paytm might separate guest and staff WiFi in their offices.

-> Here is a short, assignment-ready security policy for the co-working space.

Co-Working Space Network Security Policy

Purpose:

The purpose of this policy is to protect the co-working space network, user devices, and company data from unauthorized access and common network threats.

1. Wi-Fi Password Rules

- Use **WPA2-Enterprise or WPA3-Enterprise** for staff Wi-Fi where supported.
- Wi-Fi passwords must be strong, with at least **12 characters**, including uppercase/lowercase letters, numbers, and symbols.
- Do not share staff Wi-Fi passwords with guests or unauthorized users.
- Change passwords immediately if they are suspected to be compromised.
- Network equipment administrator passwords must be different from Wi-Fi passwords.

2. VLAN Access Restrictions

- **Marketing:** VLAN 10 – 192.168.10.0/24
- **Tech:** VLAN 20 – 192.168.20.0/24
- **Finance:** VLAN 30 – 192.168.30.0/24
- **Management:** VLAN 40 – 192.168.40.0/24
- Departments should be separated using VLANs to reduce unnecessary access between departments.
- Finance systems should only be accessible to authorized Finance users and administrators.
- Management VLAN access should be restricted to authorized IT administrators.
- Inter-VLAN communication should be allowed only when required for legitimate business purposes.

3. Guest Network

- Create a separate **Guest Wi-Fi SSID and Guest VLAN**.
- Guest users should receive **internet-only access**.

- Guests must not access Marketing, Tech, Finance, Management VLANs, printers, servers, or network-management interfaces.
- Use a guest password or captive portal and change guest credentials periodically.
- Apply reasonable bandwidth limits to prevent a few users from consuming all available bandwidth.

4. Basic Firewall Guidelines

- Use a firewall/router between the internal network and the Internet.
- **Deny incoming Internet connections by default** unless a specific service requires them.
- Allow only necessary outbound traffic and block suspicious or unauthorized traffic.
- Enable firewall logging and regularly review important security events.
- Keep the router/firewall firmware updated.
- Disable unnecessary services and remote administration from the public Internet.
- Use NAT and appropriate access-control rules to protect internal devices.

5. Security Responsibility

All users must protect their account credentials, avoid connecting unauthorized devices to the network, and report suspected security incidents to the network administrator immediately.

Policy goal: The network should follow a **least-privilege approach**, where users receive only the network access necessary for their work.